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F78AB
Actuarial and Financial Mathematics

Chapter 13: Term structure of interest rates

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Yield Curves

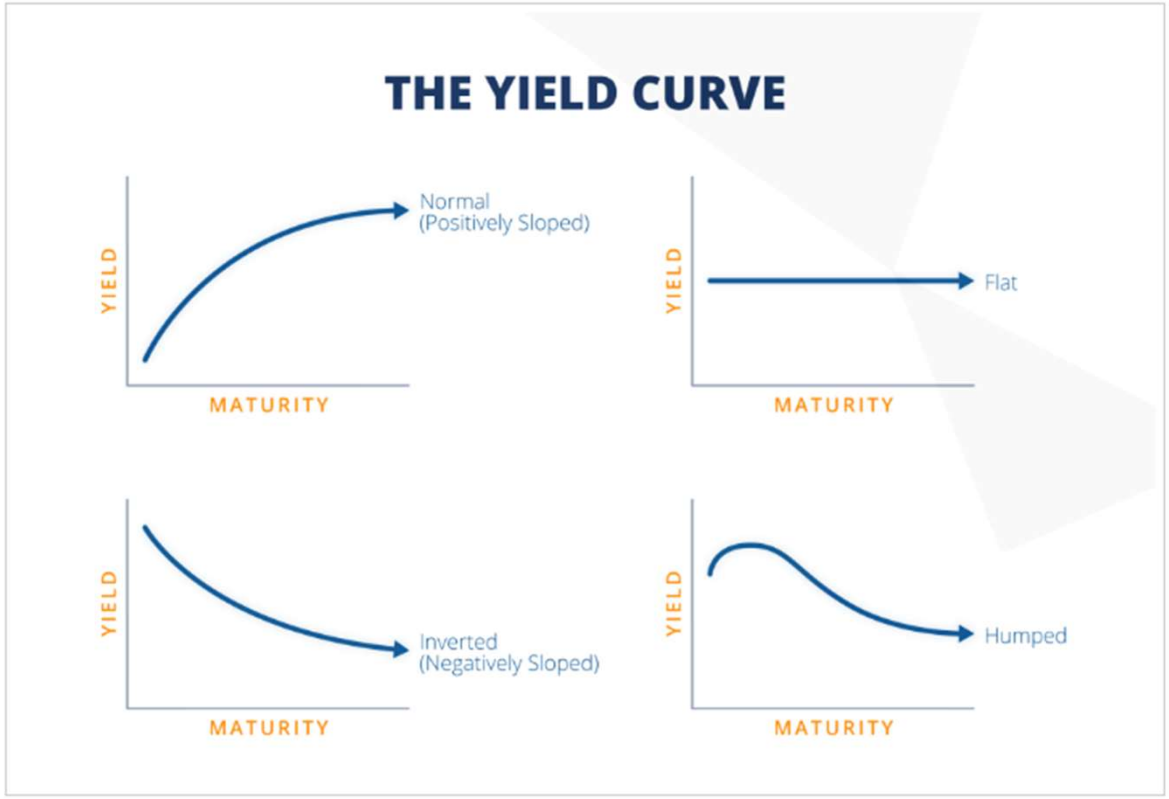
- The *term structure of interest rates* is the variation of bond yields with terms to redemption of the bonds.
- Graphically, this is called a *yield curve,*
- Yields are calculated from market prices – which are determined by demand and supply
- What factors might impact the demand and supply for bonds in the market?

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10.1 Yield Curves

- Yield curves vary from day to day. They can be:
  - Increasing (upward-sloping) – “normal”
  - Decreasing (downward-sloping) – “inverted”
  - Flat (all yields are equal)
  - Inconsistent term to redemption/yield relationship- “humped”, eg short and long terms are equal but medium term vary
- Flat and humped curves indicate uncertainty in the market. As certainty increases, the yield curve moves back towards a normal or inverted slope.

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Yield Curves

Three observed characteristics of the term structure of interest rates:

1. The change in yields of different term bonds tends to move in the same direction
2. The yields on short-term bonds are more volatile than long-term bonds
3. The yields on long-term bonds are often higher than short-term bonds

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Shape of the yield curve explained

Expectations Theory:

- Long-term interest rates reflect investors' expectations of short-term interest rates in the future
- That means that the long-term rates is made up of all the short-term rates up to that long term; ie
- The long-term rate up to time T is the geometric mean of all the intervening short-term rates up to time T

- Thus: change in yields tend to move in the same direction, and yields on short-term bonds are more volatile than long-term

Shape of the yield curve explained

Liquidity Preference Theory:

- The more risky the investment, the higher the *expected* return needs to be to encourage investors
- Long-dates bonds are more risky that short-dated bonds – when interest rates move, the price of a long-dated bond changes more than the price of a short-dated bond
- Therefore investors expected a higher return on long-dated bonfs because of the extra risk.

- Together, the Expectations Theory and Liquidity Preference Theory suggest an upward-sloping yield curve is more common than downward-sloping – as is observed in practice

Shape of the yield curve explained

Market Segmentation Theory:

- Different investors are attracted to different types of bonds
- This can be due to investment requirements, eg:
  - Pension funds attracted to long-dated indexed linked bonds to match long-term inflation linked liabilities
- Or can be due to regulatory reasons, taxation benefits, eg:
  - An investor not liable to income tax would be more interested in high-coupon bonds
- The yield for each 'segment' of the market is determined by supply and demand in that segment, eg:
  - Yield on short-term dated bonds is determined by supply and demand among short-term investors

- Market segmentation theory explains why yields at the long-end of the yield curve may decrease, however for the overall shape of the yield curve we need the Expectations Theory and the Liquidity Preference Theory

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Reminder - Zero-Coupon Bonds

- £1 nominal of a n-year **zero-coupon bond (ZCB)** pays £1 after *n* years, and nothing else
- Ie ZCB is a fixed interest security, redeemed at par in term *n* years, with **no** coupon payments.

- In UK ZCB is sometimes called a “gilt strip” – a gilt stripped of its interest payments

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Spot Rates

- The ***n-year spot rate of interest*** y_n is the annual redemption yield of an n-year ZCB.

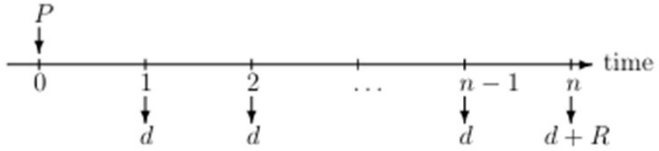
- Let P_n be the price today of an n-year ZCB. Then for $n > 0$,

- Eqn of Value: $P_n = 1 \times (1 + y_n)^{-n}$

Example: Calculate the spot rate of interest y_5 for a 5-year ZCB currently priced at 0.78

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- Consider a fixed-interest security which pays redemption proceeds of *R* at time *n*, and coupons at times *1,2,….,n-1,n*.
- If the security is priced fairly, the price, P, of this security must be equal to the price of a series of ZCBs.



$$P = d(P_1 + P_2 + \cdots + P_n) + RP_n$$
$$= d((1+y_1)^{-1} + (1+y_2)^{-2} + \ldots + (1+y_n)^{-n}) + R(1+y_n)^{-n}$$

- Otherwise, you could make money by buying a series of ZCBs and selling the fixed-interest security (or vice-versa) to make a profit – this process is called *arbitrage.*
- Thus we can consider only ZCBs

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Example

Suppose $P_1 = 0.95$ and $P_2 = 0.90$.

Calculate:

(i) the 1-year spot rate

(ii) the 2-year spot rate

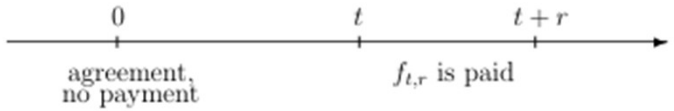
(iii) The price of £100 nominal of a 2-year fixed-interest bond with coupon rate 7%p.a., paid annually in arrears, with redemption proceeds 100%.

(iv) The gross redemption yield on the above fixed-interest bond

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Forward Rates

- The ***forward rate,*** $f_{t,r}$ is the effective annual rate of interest agreed now (ie at time 0) for an investment made at time t for a period of r years.



- The present value at time t of a unit amount paid at time $t+r$ is: $1 \; x \; (1 + f_{t,r} \;)^{-r}$

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Relationship between spot rates and forward rates

- Consider £1 invested at time 0 *for* $t+r$ years at the $(t+r)$ spot rate. It will accumulate to $\underline{(1+y_{t+r})^{t+r}}$

- Alternatively, invest the £1 for t years at the *t-year* spot rate y_t followed by investment from time t *to* $t+r$ at the forward rate $f_{t,r}$, so the £1 will accumulate to $\underline{(1+y_t)^t(1+f_{t+r})^r}$

- With no arbitrage, these must be equal, ie

$$(1+y_{t+r})^{t+r} = (1+y_t)^t(1+f_{t+r})^r$$

therefore $$(1+f_{t+r})^r = \frac{(1+y_{t+r})^{t+r}}{(1+y_t)^t}$$

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Par Yield

The ***n-year par yield*** is the coupon rate, c_n, payable annually in arrears per £1 nominal, which would give an n-year bond with coupon rate c_n a price of £1, assuming the bond is redeemed at par.

That is, for a given zero-coupon yield curve, the n-year par yield, c_n, is given by:

$$1 = c_n\,[(1+y_1)^{-1} + (1+y_2)^{-2} + \ldots\ldots + (1+y_n)^{-n}\,] + (1+y_n)^{-n}$$

$$= c_n\,[P_1 + P_2 + \ldots\ldots + P_n\,] + P_n$$

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Example: Calculate c_3 based on the following spot interest rates:

Term (in years)	Spot rate (% p.a.)
1	5
2	5.5
3	6

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Continuous-time Rates

- The ***t-year spot force of interest***, Y_t is the force of interest equivalent to the annual spot rate, y_t

$$P_t = e^{-tY^t}$$

Y_t is also called the:

- Continuously-compounded spot rate, or
- Continuous-tine spot rate

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Definition: The ***continuous-time forward rate*** $F_{t,r}$ is the force of interest equivalent to the annual rate of interest $f_{t,r}$

That is, $F_{t,r} = \ln(1 + f_{t,r})$

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THE END!!!!!!

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Attendance matters!

